USN	0	2	J	S	T	2	4	U	C	S	0	3	0	Course Code:	24CS330

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Third Semester ,B E Degree Examination

Department of Computer Science and Engineering

DATASTRUCTURES

Duration: 3 Hours | Max. Marks: 100

Instructions:

- All Questions in Part A is compulsory.
- Part B has internal choice, Answer any one full question.

PART – A

Q.NO	1	CO	1	CD	L2	PI	1.4.1	MARKS	10

Define Data structure. Explain the classification of Data structures with suitable examples.

Q.NO	2	CO	2	CD	L2	PI	2.2.4	MARKS	5+5=10

a) Demonstrate the difference between Single and doubly linked list with examples.
b) Explain the significance of header node in linked list with example.

Q.NO	3	CO	3	CD	L3	PI	3.2.2	MARKS	10

Apply expression conversion algorithm and convert the following given infix expression into prefix as well as postfix expression.(Indicate the stack content after each step)

((A – B) + C) $ (D + (E – F))

Q.NO	4	CO	4	CD	L2	PI	2.1.2	MARKS	10

Explain the following with suitable examples.

a) Complete Binary tree
b) Binary search tree
c) Balanced Binary tree
d) Skewed Binary tree

Q.NO	5	CO	5	CD	L2	PI	2.1.2	MARKS	10

Discuss the different storage structures of graphs with suitable examples.

PART – B

Q.NO	6	CO	1	CD	L3	PI	3.2.2	MARKS	5+5

a) Develop a function in C language that checks whether given two arrays are identical

or not using pointers.

b) Develop a C program using pointers to read array of n elements and display the content of the array in reverse order using pointers.

OR

Q.NO	7	CO	1	CD	L3	PI	3.2.2	MARKS	10

Define a Structure Employee with the following details name, employee ID, department name, salary and years of experience. Develop a C program to read the employee details, calculates the salary hike based on years of experience and display the details of employees of a given department.

Years of Experience	Salary hike
< 5years	5%
5 years to 10 years	10%
Above 10 years	15%

Q.NO	8	CO	2	CD	L2	PI	3.2.2	MARKS	10

Define Circular linked list and Develop C functions of circular single linked list to perform the following operations

i) count the number occurrences of key value
ii) reverse the list

OR

Q.NO	9	CO	2	CD	L3	PI	3.2.2	MARKS	10

Define Doubly linked list and Develop C functions of circular doubly linked list to perform the following operations

i) Insert at rear
ii) Search for an item.

Q.NO	10	CO	3	CD	L3	PI	4.1.2	MARKS	10

Explain the applications of Queue. Develop C functions to perform the following queue operations

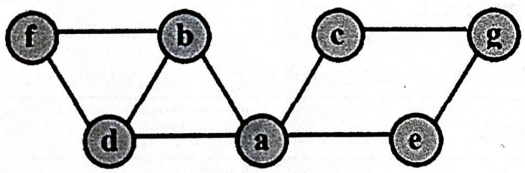
i) Enqueue or insertion
ii) Dequeue or deletion

OR

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Q.NO	11	CO	3	CD	L3	PI	4.1.2	MARKS	10
Define Tower of Hanoi problem. Provide step wise solution to solve the same with 3 discs. Develop a C function to solve Tower of Hanoi problem.									

Q.NO	12	CO	4	CD	L3	PI	4.1.2	MARKS	5+5
Develop C functions to perform the following: i) To find the minimum node in Binary Search Tree (BST). ii) Develop a C function to insert a node into Binary Search Tree.									
OR									
Q.NO	13	CO	4	CD	L3	PI	4.1.2	MARKS	5+5
Define Binary tree. Construct a binary tree by considering given traversals and also provide the post order traversal for the same. Pre-order: ABCGHDEFI In-order: GCHBDAEIF									

Q.NO	14	CO	5	CD	L3	PI	4.1.2	MARKS	5+5
a) Differentiate between i) Directed and Undirected graphs. ii) Connected and disjoint graphs b) Define the following with respect to graphs: i) Graph ii) path iii) in-degree of a vertex iv) out-degree of a vertex v) cycle									
OR									
Q.NO	15	CO	5	CD	L3	PI	4.1.2	MARKS	5+5
Construct i) BFS(Breadth First Search) tree by performing the BFS traversal ii) DFS(Depth First search) tree by performing DFS traversal from vertex A for the following graph:  Figure : Q15									

Course Outcome (CO):	
CO–1	Using the concept of pointers to arrays, structures, and functions, perform different operations on linear list with array storage representation.
CO–2	Use the concept of linked list representation to implement various operations on Linear lists with different types of linked list.
CO–3	Solve the problems of conversion and evaluation of different types of expressions using stack data structure; comprehend the implementation of different types of Queues.
CO–4	Using tree data structures solve the problems of searching, sorting, expression conversion and expression evaluation.
CO–5	Applying graph traversal techniques solve the problems of testing connectivity and acyclicity in a graph.

PI	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.1.2	Identify processes/modules/algorithms of a computer based system and parameters to solve a problem.
2.2.4	Compare and contrast alternative solution/methods to select the best methods
3.2.2	Able to produce a variety of potential design solutions suited to meet functional requirements.
4.1.2	Able to choose appropriate procedure/algorithm, dataset and test cases

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyse
L5	Evaluate
L6	Create

--- End ---